INFOSYS SE PSEUDOCODE VAULT • COMPLETE PSEUDOCODE MASTER COMPENDIUM

All 5 Core Topics Combined • 100% Solved & Traced • Total Solved Questions: 238

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Operator Precedence*

Q1. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 5
Set Integer c = 2
display a - b * c + 6
```

(A) 10
(B) 14
(C) 16
(D) 19

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** Multiplication first: 18 – (5×2) + 6 = 14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Mixed Arithmetic*

Q2. What will be the output of the following pseudocode?

```
Set Integer x = 24
Set Integer y = 7
display (x / y) * y + (x % y)
```

(A) 21
(B) 24
(C) 25
(D) 28

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** 24/7=3 and 24%7=3; 3×7+3 = 24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Logical Operators*

Q3. What will be the output of the following pseudocode?

```
Set Integer a = 12
Set Integer b = 7
Set Integer c = 5
display (a > b) and ((b - c) == 2)
```

(A) true
(B) false
(C) 2
(D) 12

Correct Answer: Option (A) : true

■ **Step-by-Step Logic:** 12>7 is true and 7–5=2 is true, so the result is true.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Logical Operators*

Q4. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x % 2 == 0) or ((x > 5) and (x < 10))
```

(A) true
(B) false
(C) 9
(D) 0

Correct Answer: Option (A) : true

■ **Step-by-Step Logic:** First condition is false, but 9>5 and 9<10 are both true.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Bitwise XOR*

Q5. What will be the output of the following pseudocode?

```
Set Integer a = 15
Set Integer b = 9
Set Integer c = 6
display (a ^ b) + c
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Bitwise AND*

Q6. What will be the output of the following pseudocode?

```
Set Integer a = 27
Set Integer b = 14
display (a & b) + (a % b)
```

(A) 19
(B) 21
(C) 23
(D) 25

Correct Answer: Option (C) : 23

■ **Step-by-Step Logic:** 27&14 = 10 and 27%14 = 13; total = 23.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *Bitwise OR*

Q7. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 11
display (a | b) - (a & b)
```

(A) 21
(B) 23
(C) 25
(D) 27

Correct Answer: Option (C) : 25

■ **Step-by-Step Logic:** 18|11=27 and 18&11=2; 27–2=25.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: *XOR + AND*

Q8. What will be the output of the following pseudocode?

```
Set Integer a = 13
Set Integer b = 7
display (a ^ b) + (a & b)
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** 13^7=10 and 13&7=5; total = 15. Infosys Medium Pseudocode Practice • Page 33

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Pre/Post Increment

Q9. What will be the output of the following pseudocode?

```
Set Integer x = 4
Set Integer y = x++ + ++x
Set Integer z = y + x
display z
```

(A) 14
(B) 15
(C) 16
(D) 17

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** x++ contributes 4 and makes x=5; ++x makes x=6 and contributes 6. y=10; z=10+6=16.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Post/Pre Decrement

Q10. What will be the output of the following pseudocode?

```
Set Integer x = 10
Set Integer y = --x
Set Integer z = y + x--
display z
```

(A) 16
(B) 17
(C) 18
(D) 19

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** --x makes x=9, y=9; x-- contributes 9, then x becomes 8. z=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Relational Logic

Q11. What will be the output of the following pseudocode?

```
Set Integer p = 8
Set Integer q = 12
display (p + q > 20) and (q - p == 4)
```

(A) true
(B) false
(C) 4
(D) 20

Correct Answer: Option (B) : false

■ **Step-by-Step Logic:** p+q=20, so 20>20 is false. The second condition is true; false AND true = false.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Ternary Logic

Q12. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 9
Set Integer m = (a % 2 == 0) ? a/2 : b*2
display m + b
```

(A) 14
(B) 16
(C) 18
(D) 21

Correct Answer: Option (B) : 16

■ **Step-by-Step Logic:** a is even, so m=7; 7+9=16.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Compound Assignment

Q13. What will be the output of the following pseudocode?

```
Set Integer x = 7
x = x * 3 - 4
x = x + (x % 5)
display x
```

(A) 18
(B) 19
(C) 20
(D) 21

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** 7×3–4=17; 17%5=2; final x=19.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Operators

Q14. What will be the output of the following pseudocode?

```
Set Integer a = 20
Set Integer b = 6
display (a / b) + (a % b) * 2
```

(A) 5
(B) 7
(C) 9
(D) 11

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** 20/6=3; 20%6=2; 3+2×2=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Left Shift

Q15. What will be the output of the following pseudocode?

```
Set Integer x = 11
Set Integer y = 2
display (x << y) - x
```

(A) 22
(B) 33
(C) 44
(D) 55

Correct Answer: Option (B) : 33

■ **Step-by-Step Logic:** 11<<2 = 44; 44–11=33.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Right Shift

Q16. What will be the output of the following pseudocode?

```
Set Integer x = 45
Set Integer y = 2
display (x >> y) + x % 4
```

(A) 10
(B) 11
(C) 12
(D) 13

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** 45>>2 = 11; 45%4=1; total=12. Infosys Medium Pseudocode Practice • Page 34

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Left Shift in Expression

Q17. What will be the output of the following pseudocode?

```
Set Integer a = 6
Set Integer b = 3
display (a << b) / 4
```

(A) 8
(B) 10
(C) 12
(D) 16

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Right Shift in Expression

Q18. What will be the output of the following pseudocode?

```
Set Integer a = 50
display (a >> 1) - (a >> 3)
```

(A) 17
(B) 19
(C) 21
(D) 23

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** 50>>1=25 and 50>>3=6; 25–6=19.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + XOR

Q19. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x << 1) ^ x
```

(A) 19
(B) 23
(C) 27
(D) 31

Correct Answer: Option (C) : 27

■ **Step-by-Step Logic:** 9<<1=18; 18 XOR 9 = 27.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + AND

Q20. What will be the output of the following pseudocode?

```
Set Integer x = 29
display (x >> 2) & 7
```

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** 29>>2=7; 7 AND 7 = 7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Double Shift

Q21. What will be the output of the following pseudocode?

```
Set Integer x = 7
Set Integer y = (x << 2)
display y >> 1
```

(A) 7
(B) 12
(C) 14
(D) 28

Correct Answer: Option (C) : 14

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (14).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift and Modulo

Q22. What will be the output of the following pseudocode?

```
Set Integer x = 37
display (x >> 1) % 6
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (A) : 0

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (0).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift Condition

Q23. What will be the output of the following pseudocode?

```
Set Integer x = 23
if ((x >> 2) >= 5)
display x - 3
else
display x + 3
end-if
```

(A) 20
(B) 23
(C) 26
(D) 28

Correct Answer: Option (A) : 20

■ **Step-by-Step Logic:** 23>>2=5, so the condition is true and 23–3=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + Arithmetic

Q24. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 2
display (a << b) - (a >> b)
```

(A) 51
(B) 53
(C) 55
(D) 57

Correct Answer: Option (B) : 53

■ **Step-by-Step Logic:** 14<<2=56; 14>>2=3; 56–3=53. Infosys Medium Pseudocode Practice • Page 35

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift Chain

Q25. What will be the output of the following pseudocode?

```
Set Integer x = 3
x = x << 2
x = x >> 1
x = x << 1
display x
```

(A) 6
(B) 10
(C) 12
(D) 24

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + OR

Q26. What will be the output of the following pseudocode?

```
Set Integer x = 5
display (x << 2) | 3
```

(A) 19
(B) 20
(C) 23
(D) 24

Correct Answer: Option (C) : 23

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (23).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + XOR

Q27. What will be the output of the following pseudocode?

```
Set Integer x = 12
display (x >> 1) ^ 5
```

(A) 1
(B) 3
(C) 5
(D) 7

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** 12>>1=6; 6 XOR 5 = 3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Bit Test with Shift

Q28. What will be the output of the following pseudocode?

```
Set Integer x = 40
display (x >> 3) & 1
```

(A) 0
(B) 1
(C) 2
(D) 5

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** 40>>3=5; 5 AND 1 = 1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shifted Difference

Q29. What will be the output of the following pseudocode?

```
Set Integer a = 31
display (a << 1) - (a >> 2)
```

(A) 54
(B) 55
(C) 56
(D) 57

Correct Answer: Option (B) : 55

■ **Step-by-Step Logic:** 31<<1=62; 31>>2=7; 62–7=55.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: For Loop with Condition

Q30. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 10 step 1 do
if (i % 3 != 0)
s = s + i
end-if
end-for
display s
```

(A) 30
(B) 37
(C) 42
(D) 45

Correct Answer: Option (B) : 37

■ **Step-by-Step Logic:** Total 1..10=55; multiples of 3 are 3+6+9=18; 55–18=37.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop

Q31. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 5 step 1 do
if (i + j == 6)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (C) : 4

■ **Step-by-Step Logic:** Pairs are (1,5),(2,4),(3,3),(4,2): 4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Sum

Q32. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to i + 1 step 1 do
s = s + j
end-for
end-for
display s
```

(A) 17
(B) 18
(C) 19
(D) 20

Correct Answer: Option (C) : 19

■ **Step-by-Step Logic:** For i=1,2,3 the inner sums are 3,6,10; total=19. Infosys Medium Pseudocode Practice • Page 36

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While Loop

Q33. What will be the output of the following pseudocode?

```
Set Integer x = 5
Set Integer s = 0
while (x <= 25)
s = s + x
x = x + 5
end-while
display s
```

(A) 50
(B) 60
(C) 75
(D) 80

Correct Answer: Option (C) : 75

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (75).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break in Nested Loop

Q34. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if (j == 3)
break
end-if
c = c + 1
end-for
end-for
display c
```

(A) 4
(B) 6
(C) 8
(D) 12

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** The inner loop adds 1 for j=1 and j=2, then breaks at j=3. 4×2=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue

Q35. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 8 step 1 do
if (i % 3 == 0)
continue
end-if
s = s + i
end-for
display s
```

(A) 24
(B) 27
(C) 30
(D) 33

Correct Answer: Option (B) : 27

■ **Step-by-Step Logic:** 1..8 sum=36; skip 3 and 6, so 36–9=27.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Reverse Loop

Q36. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 12 to 2 step -2 do
s = s + i / 2
end-for
display s
```

(A) 18
(B) 21
(C) 24
(D) 27

Correct Answer: Option (B) : 21

■ **Step-by-Step Logic:** Values of i/2: 6,5,4,3,2,1; sum=21.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Changing Step

Q37. What will be the output of the following pseudocode?

```
Set Integer i = 1
Set Integer s = 0
while (i < 20)
s = s + i
i = i * 2 + 1
end-while
display s
```

(A) 24
(B) 26
(C) 28
(D) 30

Correct Answer: Option (B) : 26

■ **Step-by-Step Logic:** i values are 1,3,7,15; sum=26.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Count

Q38. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 5 step 1 do
for j = i to 5 step 1 do
c = c + 1
end-for
end-for
display c
```

(A) 10
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Counts are 5+4+3+2+1=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Modulo

Q39. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 14 step 2 do
if (i % 4 == 0)
s = s + i
end-if
end-for
display s
```

(A) 12
(B) 16
(C) 20
(D) 24

Correct Answer: Option (D) : 24

■ **Step-by-Step Logic:** Multiples of 4 are 4,8,12; sum=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Conditional Loop

Q40. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 5 step 1 do
if (i % 2 == 1)
for j = 1 to i step 2 do
s = s + j
end-for
end-if
end-for
display s
```

(A) 12
(B) 14
(C) 16
(D) 18

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** For i=1,3,5 the inner sums are 1,4,9; total=14. Infosys Medium Pseudocode Practice • Page 37

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop State

Q41. What will be the output of the following pseudocode?

```
Set Integer x = 3
for i = 1 to 4 step 1 do
x = x + i
x = x - 1
end-for
display x
```

(A) 8
(B) 9
(C) 10
(D) 11

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Each iteration changes x by i–1. Total change=0+1+2+3=6; final x=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop Product

Q42. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 2 to 8 step 2 do
p = p * (i / 2)
end-for
display p
```

(A) 12
(B) 24
(C) 36
(D) 48

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** Factors are 1,2,3,4; product=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While + Modulo

Q43. What will be the output of the following pseudocode?

```
Set Integer n = 29
Set Integer c = 0
while (n > 0)
if (n % 2 == 1)
c = c + 1
end-if
n = n / 2
end-while
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** Binary 29 is 11101, containing four 1s.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Arithmetic

Q44. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to 3 step 1 do
s = s + i * j
end-for
end-for
display s
```

(A) 12
(B) 18
(C) 24
(D) 36

Correct Answer: Option (D) : 36

■ **Step-by-Step Logic:** The double sum factorizes: (1+2+3)×(1+2+3)=6×6=36.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break with Sum

Q45. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 12 step 2 do
s = s + i
if (s > 15)
break
end-if
end-for
display s
```

(A) 12
(B) 18
(C) 20
(D) 24

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** Running sums: 2,6,12,20; break after 20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue with Product

Q46. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 1 to 6 step 1 do
if (i % 2 == 0)
continue
end-if
p = p * i
end-for
display p
```

(A) 9
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Odd values 1,3,5 are multiplied: 1×3×5=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Shift

Q47. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 4 step 1 do
s = s + (i << 1)
end-for
display s
```

(A) 16
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (20).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Right Shift

Q48. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 8 to 32 step 8 do
s = s + (i >> 2)
end-for
display s
```

(A) 15
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** The shifted values are 2,4,6,8; sum=20. Infosys Medium Pseudocode Practice • Page 38

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop with Break

Q49. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
c = c + 1
if (c == 7)
break
end-if
end-for
end-for
display c
```

(A) 7
(B) 8
(C) 12
(D) 13

Correct Answer: Option (D) : 13

■ **Step-by-Step Logic:** The counter continues across outer iterations. It reaches 7, breaks the current inner loop, then continues; 4 more inner iterations occur, so final c=13.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop and Comparison

Q50. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 9 step 1 do
if (i * i < 50)
c = c + 1
end-if
end-for
display c
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** i=1..7 satisfy $i^2<50$; count=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Index Calculation

Q51. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 2, 9, 5]
display a[a[2]] + a[1]
```

(A) 7
(B) 9
(C) 11
(D) 13

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** a[2]=2, so a[a[2]]=a[2]=2; 2+7=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Transformation

Q52. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
for i = 0 to 3 step 1 do
a[i] = a[i] + i
end-for
display a[3] - a[1]
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (D) : 6

■ **Step-by-Step Logic:** After updates, a=[3,6,9,12]; 12−6=6.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Maximum

Q53. What will be the output of the following pseudocode?

```
Set Integer a = [8, 3, 15, 6, 11]
Set Integer m = a[0]
for i = 1 to 4 step 1 do
if (a[i] > m)
m = a[i]
end-if
end-for
display m % 4
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** Maximum is 15; 15%4=3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Sum

Q54. What will be the output of the following pseudocode?

```
Set Integer a = [2, 5, 8, 11]
Set Integer s = 0
for i = 0 to 2 step 1 do
s = s + a[i] + a[i+1]
end-for
display s
```

(A) 36
(B) 39
(C) 42
(D) 45

Correct Answer: Option (B) : 39

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (39).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Conditional Update

Q55. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 10, 13]
for i = 0 to 3 step 1 do
if (a[i] % 2 == 0)
a[i] = a[i] / 2
else
a[i] = a[i] + 1
end-if
end-for
display a[0] + a[1] + a[2] + a[3]
```

(A) 25
(B) 27
(C) 29
(D) 31

Correct Answer: Option (C) : 29

■ **Step-by-Step Logic:** Array becomes [2,8,5,14]; sum=29.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Reverse

Q56. What will be the output of the following pseudocode?

```
Set Integer a = [3, 6, 9, 12, 15]
display a[4] - a[0] + a[3] - a[1]
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 15−3+12−6=18. Infosys Medium Pseudocode Practice • Page 39

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Count

Q57. What will be the output of the following pseudocode?

```
Set Integer a = [5, 10, 15, 20, 25]
Set Integer c = 0
for i = 0 to 4 step 1 do
if (a[i] % 10 == 5)
c = c + 1
end-if
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** 5,15,25 satisfy the condition: 3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Running Sum

Q58. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8]
Set Integer s = a[0]
for i = 1 to 3 step 1 do
a[i] = a[i] + s
s = a[i]
end-for
display a[3]
```

(A) 16
(B) 20
(C) 24
(D) 30

Correct Answer: Option (B) : 20

■ **Step-by-Step Logic:** Running values become 2,6,12,20; a[3]=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Nested Index

Q59. What will be the output of the following pseudocode?

```
Set Integer a = [1, 3, 0, 2]
display a[a[3]] + a[a[1]]
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** a[3]=2, so a[2]=0; a[1]=3, so a[3]=2; total=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Difference

Q60. What will be the output of the following pseudocode?

```
Set Integer a = [12, 5, 9, 4]
Set Integer m = a[0]
for i = 1 to 3 step 1 do
if (a[i] < m)
m = a[i]
end-if
end-for
display a[0] - m
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** Minimum is 4; 12−4=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Shift

Q61. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
display (a[3] >> 1) + (a[0] << 1)
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** 9>>1=4; 3<<1=6; total=10.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array XOR

Q62. What will be the output of the following pseudocode?

```
Set Integer a = [6, 10, 12, 3]
display (a[0] ^ a[1]) + a[3]
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (15).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Comparison

Q63. What will be the output of the following pseudocode?

```
Set Integer a = [4, 9, 6, 11]
Set Integer c = 0
for i = 0 to 2 step 1 do
if (a[i] < a[i+1])
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** The true comparisons are 4<9 and 6<11, so count=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Sum with Step

Q64. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8, 10, 12]
Set Integer s = 0
for i = 0 to 5 step 2 do
s = s + a[i]
end-for
display s
```

(A) 12
(B) 16
(C) 18
(D) 20

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** Indices 0,2,4 give 2+6+10=18. Infosys Medium Pseudocode Practice • Page 40

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Rotation Logic

Q65. What will be the output of the following pseudocode?

```
Set Integer a = [10, 20, 30, 40]
Set Integer t = a[3]
for i = 3 to 1 step -1 do
a[i] = a[i-1]
end-for
a[0] = t
display a[1] + a[3]
```

(A) 30
(B) 40
(C) 50
(D) 60

Correct Answer: Option (B) : 40

■ **Step-by-Step Logic:** After right rotation: [40,10,20,30]; a[1]+a[3]=40.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length

Q66. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer n = stringLength(s)
display (n << 1) - 3
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** Length=9; 9<<1=18; 18–3=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Indexing

Q67. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[1] + s[5]
```

(A) NS
(B) NY
(C) FY
(D) SY

Correct Answer: Option (B) : NY

■ **Step-by-Step Logic:** s[1]=N and s[5]=Y, so the result is NY.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Condition

Q68. What will be the output of the following pseudocode?

```
Set String s = "COMPUTER"
if (stringLength(s) % 2 == 0)
display stringLength(s) / 2
else
display stringLength(s) + 2
end-if
```

(A) 4
(B) 5
(C) 6
(D) 10

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** COMPUTER has 8 characters; 8/2=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String + Loop

Q69. What will be the output of the following pseudocode?

```
Set String s = "CODE"
Set Integer total = 0
for i = 0 to stringLength(s)-1 step 1 do
total = total + i
end-for
display total
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (C) : 6

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (6).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Index + Shift

Q70. What will be the output of the following pseudocode?

```
Set String s = "TESTING"
Set Integer n = s[1] == 'E'
display stringLength(s) << n
```

(A) 7
(B) 14
(C) 21
(D) 28

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** The comparison is true, represented as 1; length 7<<1=14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Modulo

Q71. What will be the output of the following pseudocode?

```
Set String s = "ALGORITHM"
display stringLength(s) % 4
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** ALGORITHM has 9 characters; 9%4=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Reverse Access

Q72. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[stringLength(s)-1]
```

(A) I
(B) S
(C) Y
(D) Error

Correct Answer: Option (B) : S

■ **Step-by-Step Logic:** Last index is 6; s[6] = S. Infosys Medium Pseudocode Practice • Page 41

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Character Count

Q73. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer c = 0
for i = 0 to stringLength(s)-1 step 1 do
if (s[i] == 'E')
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** PLACEMENT contains E twice.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Right Shift

Q74. What will be the output of the following pseudocode?

```
Set String s = "INTERVIEW"
display stringLength(s) >> 1
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** INTERVIEW has 9 characters; 9>>1=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Conditional

Q75. What will be the output of the following pseudocode?

```
Set String s = "NQT2026"
if (stringLength(s) > 6)
display s[0]
else
display s[1]
end-if
```

(A) N
(B) Q
(C) T
(D) 2

Correct Answer: Option (A) : N

■ **Step-by-Step Logic:** Length is 7, so the first branch displays s[0]=N.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Arithmetic

Q76. What will be the output of the following pseudocode?

```
Integer calc(Integer a, Integer b)
a = a + b
b = a - b
return a * b
End function
display calc(7, 3)
```

(A) 60
(B) 70
(C) 80
(D) 90

Correct Answer: Option (B) : 70

■ **Step-by-Step Logic:** a becomes 10; b becomes 7; return 10×7=70.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop

Q77. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 1
for i = 1 to n step 1 do
s = s + i
end-for
return s
End function
display f(6)
```

(A) 20
(B) 21
(C) 22
(D) 23

Correct Answer: Option (C) : 22

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (22).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Shift

Q78. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n << 2) + (n >> 1)
End function
display f(10)
```

(A) 40
(B) 45
(C) 50
(D) 55

Correct Answer: Option (B) : 45

■ **Step-by-Step Logic:** 10<<2=40; 10>>1=5; total=45.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with XOR

Q79. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
return (a ^ b) * 2
End function
display f(12, 5)
```

(A) 18
(B) 26
(C) 30
(D) 34

Correct Answer: Option (A) : 18

■ **Step-by-Step Logic:** 12 XOR 5 = 9; 9×2=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Condition

Q80. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n % 4 == 0)
return n / 2
else
return n + 4
end-if
End function
display f(14)
```

(A) 9
(B) 14
(C) 18
(D) 28

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 14 is not divisible by 4, so 14+4=18. Infosys Medium Pseudocode Practice • Page 42

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Digit Logic

Q81. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
while (n > 0)
s = s + n % 10
n = n / 10
end-while
return s
End function
display f(386)
```

(A) 14
(B) 15
(C) 16
(D) 17

Correct Answer: Option (D) : 17

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (17).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Nested

Q82. What will be the output of the following pseudocode?

```
Integer g(Integer x)
return x * 2
End function
Integer f(Integer x)
return g(x) + g(x+1)
End function
display f(4)
```

(A) 16
(B) 18
(C) 20
(D) 22

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** g(4)=8 and g(5)=10; total=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Modulo

Q83. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n * n + 3) % 10
End function
display f(7)
```

(A) 2
(B) 5
(C) 8
(D) 9

Correct Answer: Option (A) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (A) (2).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Loop Product

Q84. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer p = 1
for i = 1 to n step 1 do
if (i % 2 == 0)
p = p * i
end-if
end-for
return p
End function
display f(6)
```

(A) 24
(B) 36
(C) 48
(D) 60

Correct Answer: Option (C) : 48

■ **Step-by-Step Logic:** Multiply even values 2×4×6=48.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Boolean

Q85. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if ((a & 1) == 1 and (b & 1) == 0)
return a + b
else
return a - b
end-if
End function
display f(9, 6)
```

(A) 3
(B) 9
(C) 15
(D) 54

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** 9 is odd and 6 is even, so return 9+6=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Array Formula

Q86. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b, Integer c)
return (a + b) ^ c
End function
display f(5, 3, 6)
```

(A) 4
(B) 8
(C) 12
(D) 14

Correct Answer: Option (D) : 14

■ **Step-by-Step Logic:** 5+3=8; 8 XOR 6 = 14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Recursive-style

Q87. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n <= 1)
return 1
end-if
return n + f(n-2)
End function
display f(7)
```

(A) 10
(B) 12
(C) 16
(D) 18

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (16).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Shifted Return

Q88. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer a = n << 1
Set Integer b = n >> 1
return a - b
End function
display f(13)
```

(A) 13
(B) 19
(C) 20
(D) 21

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** 13<<1=26; 13>>1=6; difference=20. Infosys Medium Pseudocode Practice • Page 43

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Multiple Updates

Q89. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
a = a * 2
b = b + a
a = a - b
return a + b
End function
display f(4, 3)
```

(A) 7
(B) 8
(C) 9
(D) 10

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** a=8; b=11; a=-3; return -3+11=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop and Shift

Q90. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
for i = 1 to n step 1 do
s = s + (i << 1)
end-for
return s
End function
display f(5)
```

(A) 20
(B) 25
(C) 30
(D) 35

Correct Answer: Option (C) : 30

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (30).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Function

Q91. What will be the output of the following pseudocode?

```
Integer f(Integer x)
return (x >> 1) + 3
End function
Set Integer a = [8, 12, 16]
display f(a[2])
```

(A) 9
(B) 10
(C) 11
(D) 12

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** a[2]=16; 16>>1=8; 8+3=11.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + XOR

Q92. What will be the output of the following pseudocode?

```
Set Integer x = 0
for i = 1 to 5 step 1 do
x = x ^ i
end-for
display x
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (A) : 0

■ **Step-by-Step Logic:** 0^1=1, ^2=3, ^3=0, ^4=4, ^5=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + AND

Q93. What will be the output of the following pseudocode?

```
Set Integer x = 15
for i = 1 to 3 step 1 do
x = x & (14 - i)
end-for
display x
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** x=15&13=13; 13&12=12; 12&11=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Condition + Shift

Q94. What will be the output of the following pseudocode?

```
Set Integer x = 19
if ((x & 1) == 1)
x = x << 1
x = x + 3
else
x = x >> 1
end-if
display x
```

(A) 19
(B) 38
(C) 41
(D) 44

Correct Answer: Option (C) : 41

■ **Step-by-Step Logic:** 19 is odd; 19<<1=38; +3=41.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Loop + Modulo

Q95. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if ((i + j) % 3 == 0)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (B) : 5

■ **Step-by-Step Logic:** Pairs with sum divisible by 3: 5 pairs. Correct answer B.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Right Shift

Q96. What will be the output of the following pseudocode?

```
Set Integer a = [18, 25, 32, 41]
display (a[3] >> 2) + (a[0] % 5)
```

(A) 11
(B) 12
(C) 13
(D) 14

Correct Answer: Option (C) : 13

■ **Step-by-Step Logic:** 41>>2=10; 18%5=3; total=13. Infosys Medium Pseudocode Practice • Page 44

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Function + Logical

Q97. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if (a > 10 and b < 10)
return (a << 1) + b
else
return a + (b << 1)
end-if
End function
display f(12, 7)
```

(A) 24
(B) 26
(C) 31
(D) 38

Correct Answer: Option (C) : 31

■ **Step-by-Step Logic:** 12>10 and 7<10, so (12<<1)+7 = 24+7=31.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + Array Update

Q98. What will be the output of the following pseudocode?

```
Set Integer a = [1, 2, 3, 4, 5]
for i = 1 to 4 step 1 do
a[i] = a[i] + a[i-1]
end-for
display a[4]
```

(A) 10
(B) 12
(C) 15
(D) 16

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Running cumulative values become [1,3,6,10,15].

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Function + Modulo + Shift

Q99. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return ((n % 5) << 2) + (n >> 3)
End function
display f(26)
```

(A) 5
(B) 7
(C) 9
(D) 11

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** 26%5=1; 1<<2=4; 26>>3=3; total=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Conditions

Q100. What will be the output of the following pseudocode?

```
Set Integer a = 9
Set Integer b = 6
Set Integer c = 3
if (a > b)
if ((a - b) == c)
display a + b + c
else
display a - b
end-if
else
display b + c
end-if
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 9>6 and 9−6=3, so the nested condition is true; 9+6+3=18.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q101. How many times does the loop body run, and

```
what is printed?
Integer
i, count
Set
i = 1, count = 0
while
i < 20 do
i
= i * 2
count
= count + 1
end
while
Print
count
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** Trace i and count: start i=1. Pass 1: i=2, count=1. Pass 2: i=4, count=2. Pass 3: i=8, count=3. Pass 4: i=16, count=4. Pass 5: i=32, count=5. Now i=32 is not less than 20, so the loop stops. Printed value is 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q102. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code? Integer division is used.
Integer
a, b
Set
a = 17, b = 5
Print
(a / b) + (a % b)
```

(A) 3
(B) 4
(C) 5
(D) 7

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** Integer division: 17 / 5 = 3. Remainder: 17 % 5 = 2. Printed value = 3 + 2 = 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q103. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code?
Integer
x, i
Set
x = 0
for
i = 1 to 6 do
if
i % 2 == 0 then
continue
end
if
x
= x + i
end
for
Print
x
```

(A) 9
(B) 12
(C) 16
(D) 21

Correct Answer: Option (A) : 9

■ **Step-by-Step Logic:** continue skips even values of i. The loop runs i = 1 to 6, so only the odd values 1, 3 and 5 are added. x = 1 + 3 + 5 = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q104. What will be the output of the following pseudocode?

```
What value is printed by the following
Pseudo Code?
Function
Mystery(n)
if
n <= 1 then
return
1
end
if
return
n * Mystery(n - 1)
End
Function
Print
Mystery(5)
```

(A) 15
(B) 24
(C) 120
(D) 125

Correct Answer: Option (C) : 120

■ **Step-by-Step Logic:** Mystery(n) computes n!. Mystery(5) = 5 × 4 × 3 × 2 × 1 = 120.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q105. What will be the output of the following pseudocode?

```
What is printed after the following Pseudo
Code runs?
Integer
a, b
Set
a = 12, b = 8
a
= a + b
b
= a - b
a
= a - b
Print
a, b
```

(A) 12 8
(B) 8 12
(C) 20 8
(D) 20 12

Correct Answer: Option (B) : 8 12

■ **Step-by-Step Logic:** This swaps the two values without a third variable. a = 12+8 = 20, b = 20-8 = 12, a = 20-12 = 8. Output: 8 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q106. What will be the output of the following pseudocode?

```
What is printed by the following Pseudo
Code?
Integer
i, j, count
Set
count = 0
for
i = 1 to 4 do
for
j = 1 to i do
count
= count + 1
end
for
end
for
Print
count
```

(A) 4
(B) 8
(C) 10
(D) 16

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** The inner loop runs i times for each i from 1 to 4. Total iterations = 1+2+3+4 = 10. So count prints 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q107. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code?
Integer
a, b
Set
a = 9, b = 5
Print
a ^ b
```

(A) 4
(B) 12
(C) 13
(D) 14

Correct Answer: Option (B) : 12

■ **Step-by-Step Logic:** 9 in binary is 1001 and 5 is 0101. XOR sets a bit where the bits differ: 1100, which is 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q108. What will be the output of the following pseudocode?

```
What value is printed by the following
Pseudo Code?
Function
F(n)
if
n == 0 then return 0
if
n == 1 then return 1
return
F(n - 1) + F(n - 2)
End
Function
Print
F(6)
```

(A) 5
(B) 8
(C) 13
(D) 21

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** F is the Fibonacci sequence: F(0)=0, F(1)=1, F(2)=1, F(3)=2, F(4)=3, F(5)=5, F(6)=8. So the printed value is 8.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q109. What will be the output of the following pseudocode?

```
A queue follows FIFO. What is printed by the
following Pseudo Code?
Queue
Q
enqueue(Q,
2)
enqueue(Q,
4)
enqueue(Q,
6)
dequeue(Q)
enqueue(Q,
8)
Print
front(Q)
```

(A) 2
(B) 4
(C) 6
(D) 8

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** After three enqueues the queue is [2, 4, 6] with 2 at the front. dequeue removes 2. enqueue 8 makes [4, 6, 8]. front returns 4.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q110. What will be the output of the following pseudocode?

```
A stack follows LIFO. What is printed by the
following Pseudo Code?
Stack
S
push(S,
5)
push(S,
8)
push(S,
3)
pop(S)
push(S,
7)
Print
peek(S)
```

(A) 3
(B) 5
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Stack after the three pushes: [5, 8, 3] with 3 on top. pop removes 3. push 7 makes the stack [5, 8, 7]. peek returns the top element, which is 7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q111. What will be the output of the following pseudocode?

```
What is printed?
Integer
n, d
Set
n = 48, d = 0
while
n > 1 do
n
= n / 2
d
= d + 1
end
while
Print
d
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** 48→24→12→6→3→1. That is five divisions. d = 5. (Integer division: 3/2=1.)

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q112. What will be the output of the following pseudocode?

```
What is printed?
Integer
A[5] = {2, 4, 6, 8, 10}
Integer
i, s
Set
s = 0
for
i = 0 to 4 do
if
A[i] % 4 == 0 then s = s + A[i]
end
for
Print
s
```

(A) 12
(B) 18
(C) 24
(D) 30

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Values divisible by 4 are 4 and 8. s = 4 + 8 = 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q113. What will be the output of the following pseudocode?

```
What is printed?
Integer
x
Set
x = 10
Print
x & 3
```

(A) 1
(B) 2
(C) 3
(D) 8

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** 10 is 1010 and 3 is 0011. AND is 0010, which is 2.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q114. What will be the output of the following pseudocode?

```
What is printed?
Stack
S
push(S,
1)
push(S,
4)
push(S,
9)
pop(S)
pop(S)
Print
peek(S)
```

(A) 1
(B) 4
(C) 9
(D) Empty

Correct Answer: Option (A) : 1

■ **Step-by-Step Logic:** After three pushes the top is 9. Two pops remove 9 then 4. peek returns 1.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q115. What will be the output of the following pseudocode?

```
What does G(4) return?
Function
G(n)
if
n <= 1 then return 1
return
n * G(n - 1)
End
Function
Print
G(4)
```

(A) 10
(B) 16
(C) 24
(D) 64

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** G is factorial. G(4)=4×3×2×1=24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q116. What will be the output of the following pseudocode?

```
[Bitwise]
What is printed?
Integer
a
Set
a = 6
Print
a | 3
```

(A) 7
(B) 5
(C) 2
(D) 18

Correct Answer: Option (A) : 7

■ **Step-by-Step Logic:** 6 is 0110, 3 is 0011, OR is 0111 = 7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q117. What will be the output of the following pseudocode?

```
[Queue]
What is printed?
Queue
Q
enqueue(Q,
11)
enqueue(Q,
22)
enqueue(Q,
33)
dequeue(Q)
dequeue(Q)
Print
front(Q)
```

(A) 11
(B) 22
(C) 33
(D) Empty

Correct Answer: Option (C) : 33

■ **Step-by-Step Logic:** Queue starts [11,22,33]. Two dequeues remove 11 then 22. Front is 33.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q118. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, s
Set
s = 0
for
i = 1 to 10 do
if
i % 2 == 1 then s = s + i
end
for
Print
s
```

(A) 25
(B) 30
(C) 45
(D) 55

Correct Answer: Option (A) : 25

■ **Step-by-Step Logic:** The loop adds the odd values 1+3+5+7+9 = 25.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q119. What will be the output of the following pseudocode?

```
[Recursion]
What does S(5) return?
Function
S(n)
if
n == 0 then return 0
return
n + S(n - 1)
End
Function
Print
S(5)
```

(A) 10
(B) 15
(C) 16
(D) 25

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** S(n) is the sum 5+4+3+2+1+0 = 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q120. What will be the output of the following pseudocode?

```
[Arrays]
What is printed?
Integer
A[4] = {3, 5, 7, 9}
Integer
i, p
Set
p = 1
for
i = 0 to 3 do
p
= p * A[i]
end
for
Print
p
```

(A) 24
(B) 315
(C) 945
(D) 105

Correct Answer: Option (C) : 945

■ **Step-by-Step Logic:** Product 3×5×7×9 = 945.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q121. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, p
Set
p = 1
for
i = 1 to 4 do
p
= p * i
end
for
Print
p
```

(A) 10
(B) 16
(C) 24
(D) 120

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** p multiplies 1×2×3×4 = 24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q122. What will be the output of the following pseudocode?

```
[Data
Structures] Queue Q receives 5, then 15, then 25. One
dequeue
runs. What is front(Q)?
Queue
Q
enqueue(Q,
5)
enqueue(Q,
15)
enqueue(Q,
25)
dequeue(Q)
Print
front(Q)
```

(A) 5
(B) 15
(C) 25
(D) Empty

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** FIFO: 5 leaves. Front is 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q123. What will be the output of the following pseudocode?

```
[Data
Structures] After the operations, what does peek return?
Stack
S
push(S,
10)
push(S,
20)
push(S,
30)
pop(S)
push(S,
40)
Print
peek(S)
```

(A) 10
(B) 20
(C) 40
(D) 30

Correct Answer: Option (C) : 40

■ **Step-by-Step Logic:** Stack after pops/pushes: [10, 20, 40]. peek = 40.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q124. What will be the output of the following pseudocode?

```
[Bitwise]
What is printed?
Integer
a, b
Set
a = 13, b = 6
Print
a & b
```

(A) 4
(B) 5
(C) 7
(D) 11

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** 13 is 1101, 6 is 0110, AND is 0100 = 4.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q125. What will be the output of the following pseudocode?

```
[Recursion]
What does P(4) return?
Function
P(n)
if
n == 1 then return 2
return
P(n - 1) * 2
End
Function
Print
P(4)
```

(A) 8
(B) 12
(C) 16
(D) 24

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** P(1)=2, P(2)=4, P(3)=8, P(4)=16.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q126. What will be the output of the following pseudocode?

```
What
does R(6) return?
Function
R(n)
if
n < 3 then return n
return
R(n - 1) + 2
End
Function
Print
R(6)
```

(A) 6
(B) 8
(C) 10
(D) 12

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** R(2)=2. Each later call adds 2. R(6)=2 + 2×4 = 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q127. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
3)
push(S,
6)
push(S,
9)
pop(S)
push(S,
12)
Print
peek(S)
```

(A) 3
(B) 6
(C) 12
(D) 9

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** After pop, 9 is gone. 12 is pushed. peek returns 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q128. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, c
Set
c = 0
for
i = 1 to 8 do
if
i % 2 == 0 then c = c + i
end
for
Print
c
```

(A) 16
(B) 18
(C) 20
(D) 36

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Even values 2+4+6+8 = 20.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q129. What will be the output of the following pseudocode?

```
What
is printed?
Integer
x
Set
x = 11
Print
x | 4
```

(A) 15
(B) 7
(C) 8
(D) 44

Correct Answer: Option (A) : 15

■ **Step-by-Step Logic:** 11 is 1011, 4 is 0100, OR is 1111 = 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q130. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
7)
enqueue(Q,
14)
enqueue(Q,
21)
dequeue(Q)
enqueue(Q,
28)
Print
front(Q)
```

(A) 7
(B) 14
(C) 21
(D) 28

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** 7 is removed first. The new front is 14.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q131. What will be the output of the following pseudocode?

```
What
does U(4) return?
Function
U(n)
if
n == 0 then return 1
return
n + U(n - 1)
End
Function
Print
U(4)
```

(A) 8
(B) 9
(C) 11
(D) 24

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** U(0)=1, then +1+2+3+4 = 11.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q132. What will be the output of the following pseudocode?

```
What
is printed?
Integer
z
Set
z = 14
Print
z & 7
```

(A) 6
(B) 7
(C) 8
(D) 21

Correct Answer: Option (A) : 6

■ **Step-by-Step Logic:** 14 is 1110, 7 is 0111, AND is 0110 = 6.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q133. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, n
Set
n = 0
for
i = 1 to 6 do
if
i % 3 == 0 then n = n + i
end
for
Print
n
```

(A) 6
(B) 9
(C) 12
(D) 21

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Multiples of 3 from 1 to 6 are 3 and 6. Sum = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q134. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
4)
enqueue(Q,
8)
enqueue(Q,
12)
dequeue(Q)
Print
front(Q)
```

(A) 4
(B) 8
(C) 12
(D) Empty

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** 4 leaves first. Front is 8.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q135. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
2)
push(S,
5)
push(S,
8)
pop(S)
push(S,
11)
Print
peek(S)
```

(A) 2
(B) 5
(C) 11
(D) 8

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** Pop removes 8. Push 11. peek returns 11.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q136. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
6)
push(S,
9)
push(S,
15)
pop(S)
push(S,
21)
Print
peek(S)
```

(A) 6
(B) 9
(C) 21
(D) 15

Correct Answer: Option (C) : 21

■ **Step-by-Step Logic:** Pop removes 15. Push 21. peek returns 21.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q137. What will be the output of the following pseudocode?

```
What
is printed?
Integer
k
Set
k = 21
Print
k >> 2
```

(A) 5
(B) 6
(C) 10
(D) 84

Correct Answer: Option (A) : 5

■ **Step-by-Step Logic:** 21 is 10101. Right shift 2 gives 101 = 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q138. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, t
Set
t = 1
for
i = 1 to 5 do
t
= t * 2
end
for
Print
t
```

(A) 10
(B) 16
(C) 32
(D) 64

Correct Answer: Option (C) : 32

■ **Step-by-Step Logic:** Start 1, multiply by 2 five times: 32.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q139. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
3)
enqueue(Q,
9)
enqueue(Q,
15)
dequeue(Q)
Print
front(Q)
```

(A) 3
(B) 9
(C) 15
(D) Empty

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** 3 leaves. Front is 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q140. What will be the output of the following pseudocode?

```
What
does W(5) return?
Function
W(n)
if
n <= 1 then return 2
return
W(n - 1) + 3
End
Function
Print
W(5)
```

(A) 11
(B) 14
(C) 15
(D) 17

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** W(1)=2. Add 3 four times: 2+12=14.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q141. What will be the output of the following pseudocode?

```
What will be the
output of the following pseudo code
Integer
i, j, k, d
Set
j = 6, i = 8, k = 2, d = 1
if
((j ^ i) mod k equals 0) then
d = d + 1
else
d = d - 1
end
if
Print
d
```

(A) 1
(B) 3
(C) 0
(D) 2

Correct Answer: Option (D) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (2).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q142. What will be the output of the following pseudocode?

```
1.     #include<iostream>
using namespace std;
class Test  {
static  int x;
public:
Test()   {    x++;    }
static int getX()  {   return x;    }   };
int Test::x=0;
int main()   {
cout<< Test::getX()<<“ “;
Test t[5];

cout<< Test::getX();      }
```

(A) 0 0
(B) 5 5
(C) 0 5
(D) a) Run time error

Correct Answer: Option (C) : 0 5

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (0 5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q143. What will be the output of the following pseudocode?

```
1.     #include<iostream>
using namespace std;
class Point    {
Point()
 {  cout<<“Constructor called”;   }
};
int main(){
Point t1;
return 0;
}
```

(A) Constructor called
(B) Compile time error
(C) Run time error
(D) Blank line

Correct Answer: Option (B) : Compile time error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compile time error).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q144. What will be the output of the following pseudocode?

```
1.     #include<stdio.h>
using namespace std;
int main()
{
for (int x = 10; x >= 0; x--)
 {
 int z = x & (x >> 1);
 if (z)
 printf("%d ", x);
 }

 }
```

(A) 7 6 3
(B) 7 5 4
(C) 6 7 8
(D) 6 5 4

Correct Answer: Option (A) : 7 6 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (7 6 3).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q145. What will be the output of the following pseudocode?

```
1.     Where are placed the list of processes that are
prepared to be executed and waiting?
```

(A) Job queue
(B) Ready queue
(C) Execution queue
(D) Process queue

Correct Answer: Option (B) : Ready queue

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (Ready queue).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q146. What will be the output of the following pseudocode?

```
1.     Who among the following can block the running
process?
```

(A) Fork
(B) Read
(C) Down
(D) All of these

Correct Answer: Option (D) : All of these

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (All of these).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q147. What will be the output of the following pseudocode?

```
1.     Which of the following does not interrupt the
running process?
```

(A) Timer interrupt
(B) Device
(C) Power failure
(D) Scheduler process

Correct Answer: Option (D) : Scheduler process

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (Scheduler process).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q148. What will be the output of the following pseudocode?

```
1.    The \u0021 article referred to as a
```

(A) Unicode escape sequence
(B) Octal escape
(C) Hexadecimal
(D) Line feed

Correct Answer: Option (A) : Unicode escape sequence

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (Unicode escape sequence).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q149. What will be the output of the following pseudocode?

```
1.    Which of the following is a valid declaration of a
char?
```

(A) char ch = '\utea';
(B) char ca = 'tea';
(C) char cr = \u0223;
(D) char cc = '\itea';

Correct Answer: Option (A) : char ch = '\utea';

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (char ch = '\utea';).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q150. What will be the output of the following pseudocode?

```
1.    Which of the following is a valid long literal?
```

(A) ABH8097
(B) L990023
(C) 904423
(D) 0xnf029L

Correct Answer: Option (D) : 0xnf029L

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (0xnf029L).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q151. What will be the output of the following pseudocode?

```
1.    What does the expression float a = 35 / 0 return?
```

(A) 0
(B) Not a Number
(C) Infinity
(D) Run time exception

Correct Answer: Option (C) : Infinity

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (Infinity).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q152. What will be the output of the following pseudocode?

```
1.    Which option is false about the final keyword?
a)   A) A final method cannot be overridden in its
subclasses.
b)   B) A final class cannot be extended.
c)   C) A final class cannot extend other classes.
d)   D) A final method can be inherited.
```

(A) A&B
(B) B&C
(C) C&D
(D) None of these

Correct Answer: Option (C) : C&D

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (C) (C&D).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q153. What will be the output of the following pseudocode?

```
1.    Construct post order Transversal for the
constructed tree with the given In-Order and Preorder
Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

(A) 9 10 7 8 6
(B) 9 7 6 10 8
(C) 8 9 10 7 6
(D) 9 8 10 7 6

Correct Answer: Option (A) : 9 10 7 8 6

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (9 10 7 8 6).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q154. 1. What will be the output of this program fragment?

```
int i = 263;
      putchar(i);
```

(A) prints 263
(B) prints the ASCII equivalent of 263
(C) rings the bell
(D) prints garbage

Correct Answer: Option (C) : rings the bell

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (C) (rings the bell).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q155. 1. What will be the output of this statement?

```
printf (“%d”, 9/5);
```

(A) 1.8
(B) 1.0
(C) 2.0
(D) None of these

Correct Answer: Option (D) : None of these

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (None of these).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q156. 1. What will be the output of this program?

```
#include<stdio.h>
      int main() {
      int i = 5;
      printf("%d", i = ++i == 6);
      return 0;
      }
```

(A) 2
(B) 6
(C) 4
(D) 1

Correct Answer: Option (D) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (1).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q157. What will be the output of the following pseudocode?

```
1.     Which of the following declarations is invalid in
C language?
```

(A) char
(B) char str[] = "campus monk is the best platform for learn";
(C) char str[20] = "campus monk is the best platform for learn";
(D) char[] str = "campus monk is the best platform for learn";

Correct Answer: Option (D) : char[] str = "campus monk is the best platform for learn";

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (char[] str = "campus monk is the best platform for learn";).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q158. 1. What will be the output of this program?

```
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf("%d", h);
   return 0;
}
```

(A) 1
(B) error: redeclaration of an enumerator
(C) h
(D) 3

Correct Answer: Option (B) : error: redeclaration of an enumerator

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (error: redeclaration of an enumerator).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q159. 1. What will be the output of this program?

```
main ()
{
 if (5 < '5')
        printf("5")
 else
        printf(" NOT EQUAL TO 5.")

}
```

(A) ENQ
(B) 5
(C) I
(D) Not equal to 5

Correct Answer: Option (B) : 5

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q160. 1. What will be the output of the following pseudo code?

```
Input m = 9,n = 6,
m= m + 1;
n= n - 1;
m= m + n;
if(m>n)
print m
else
print n
```

(A) 5
(B) 7
(C) 9
(D) 12

Correct Answer: Option (A) : 5

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q161. 1. What will be the output of the following pseudocode?

```
Integer i
Set i = 3
do
print i + 3
i = i - 1
while(i not equals 0)
end while
```

(A) 6 6 6
(B) 6 5 6
(C) 5 5 5
(D) 6 5 4

Correct Answer: Option (D) : 6 5 4

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (D) (6 5 4).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q162. What will be the output of the following pseudocode?

```
1.     If the sequence of operations are performed on a
stack, the sequence of pop will be?

Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(),
Pop(), Push(2), Pop().
```

(A) 2 2 1 1 2
(B) 2 2 1 2 2
(C) 2 1 2 2 1
(D) 2 1 2 2 2

Correct Answer: Option (A) : 2 2 1 1 2

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (2 2 1 1 2).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q163. What will be the output of the following pseudocode?

```
1.     In an empty BST, insert the following elements in
a similar sequence. Which element will be at lowest
level?

105     95     122     99     97    120
```

(A) 95
(B) 97
(C) 99
(D) 120

Correct Answer: Option (B) : 97

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (97).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q164. What will be the output of the following pseudocode?

```
T The concatenation of two lists is to be performed in
O(1) time. Which of the following implementation of a
list should be used?
```

(A) a) Singly linked list
(B) a) Doubly linked list
(C) a) Circular doubly linked list
(D) a) Array implementation of list

Correct Answer: Option (C) : a) Circular doubly linked list

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (a) Circular doubly linked list).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q165. What will be the output of the following pseudocode?

```
1.     Which of the following sequence can be obtained in
the output (in same order) using a stack assuming that
the input is in the sequence 1,2,3,4,5 order?
```

(A) a) 3,4,5,1,2
(B) a) 3,4,5,2,1
(C) a) 1,5,2,3,4
(D) a) 5,4,3,1,2

Correct Answer: Option (B) : a) 3,4,5,2,1

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (a) 3,4,5,2,1).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q166. What will be the output of the following pseudocode?

```
1.     Let A be a square matrix of size nxn. Consider the
following program and state the output.
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do

Output (A[i][j]);
```

(A) a) The matrix A itself
(B) a) Transpose of matrix A
(C) a) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
(D) a) None of the above

Correct Answer: Option (A) : a) The matrix A itself

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (a) The matrix A itself).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q167. What will be the output of the following pseudocode?

```
1.     The value of j at the end of the execution of the
following pseudo code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)

end for
```

(A) 10
(B) 4
(C) 6
(D) 7

Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (A) (10).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q168. What will be the output of the following pseudocode?

```
1.     What is printed by the print statement in the
program P1 assuming call by reference parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z

}
```

(A) 10, 3
(B) 31, 3
(C) 27, 7
(D) None of these

Correct Answer: Option (B) : 31, 3

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (31, 3).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q169. What will be the output of the following pseudocode?

```
1.     The value returned by the following function for
f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);

}
```

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (7).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q170. What will be the output of the following pseudocode?

```
1.     Find the output of the following pseudo code.
Integer n
Set n = 11
Integer arr[4] = { 10, 11, 1, 2 }
arr[1]=arr[1]-arr[3]
arr[2]=arr[2]+arr[2]
arr[3]=1+arr[3]
n=arr[3]
if(n>arr[2])
n=n-1
Else
n=n-1
End if

Print n
```

(A) 7
(B) 18
(C) 22
(D) 2

Correct Answer: Option (D) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (2).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q171. What will be the output of the following pseudocode?

```
1.     Find the output of the following pseudo code.
Integer x,y,z
Set x=9, y=2, z=1
if ( x+y > 8)
if ( 7> x+z)
z=7
End if
If ( 9> x-z)
z=9
End if
y=y+z+1
End if

Print x+y+z
```

(A) 31
(B) 11
(C) 42
(D) 30

Correct Answer: Option (D) : 30

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (30).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q172. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c >
b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c <
a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a
Print a + b + c
```

(A) 71
(B) 56
(C) 47
(D) 54

Correct Answer: Option (D) : 54

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (54).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q173. How many times campus monk will be printed? Set k = 1, p = 4 Int j = p + k / 2 If (j equals p) Print "campus monk" End if j = p + k / 5 If (j not equal p) Print "campus monk" End if Print "campus monk"

(A) 1 Times
(B) 2 Times
(C) 3 Times
(D) 4 Times

Correct Answer: Option (B) : 2 Times

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (2 Times).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q174. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h> int main(void) { int a = 10, x; x = a-- + +445; printf ("%d", x); return 0; }
```

(A) 19
(B) 435
(C) 455
(D) 23

Correct Answer: Option (C) : 455

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (455).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q175. What will be the output of the following pseudocode?

```
Int p, q, r Set p = 6, q = 4, r = 7 p = (r + 5) + p If ((q & r & p) < (p & 6)) { r = 3 ^ r q = q & q } Print p + q + r
```

(A) 44
(B) 41
(C) 11
(D) 26

Correct Answer: Option (D) : 26

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (26).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q176. What will be the output of the following pseudocode?

```
What is the output of this code? Int p, q, r; Set p = 0, q = 7, r = 9; If ((q + p) < (p + q) && (p < r)) { p = 11 + q; } Print p + q + r;
```

(A) 12
(B) 16
(C) 27
(D) 10

Correct Answer: Option (B) : 16

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (16).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q177. What will be the output of the following pseudocode? Initial values: a = 4, b = 4, c = 7 Int fun(int a, int b, int c) { For (each c from 3 to 5) { a = (c + c) ^ b; If ((a + c) < (c + a)) { b = (a + 11) + c; } Else { c = b + b; a = 3 + b; continue; } } Return a + b; }

(A) 10
(B) 12
(C) Error
(D) 11

Correct Answer: Option (D) : 11

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (11).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q178. What will be the output of this pseudocode? a = 7, b = 2, c = 8 If ((b ^ a) < a && (2 ^ 7) < c) a = b + c Else c = (4 + 12) + a b = 11 + a End if Print (a + b + c)

(A) 19
(B) 21
(C) 20
(D) 22

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (20).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q179. What will be the output of the following pseudocode?

```
Find the output of the code snippet. Integer x, y, z Set x = 1, y = 8, z = 2 If (x ^ y && x ^ y && x ^ y ^ z) x = 1 End if Print x + y + z
```

(A) 11
(B) 13
(C) 16
(D) 9

Correct Answer: Option (A) : 11

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (11).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q180. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h> void main() { int a = 10; if (a == 010) printf("BCA"); else printf("B.Tech"); }
```

(A) BCA
(B) B.tech
(C) 10
(D) Error

Correct Answer: Option (B) : B.tech

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (B.tech).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q181. What will be the output of the following pseudocode?

```
Find the output of the following code. #include <stdio.h>
void main () { int x = 91; x >>= 2; printf ("%d\n", x); }
```

(A) 182
(B) 91
(C) 22
(D) 82

Correct Answer: Option (C) : 22

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (22).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q182. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { int num = 987; int rem; while(num
!= 0) { rem = num % 4; num = num / 10; } printf("%d",
rem); }
```

(A) 0
(B) 1
(C) 10
(D) 5

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (1).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q183. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { float x = 0.0; long int y = 10;
printf("%d", sizeof(y) == sizeof(x + y)); return 0; }
```

(A) 3
(B) 1
(C) 5
(D) 2

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q184. What will be the output of the following pseudocode?

```
Predict the output of the given pseudocode if the value
of the number is 6: Read number k = 2 i = 2 while i <=
number: k = k * i i = i + 1 end while write k
```

(A) 1440.0
(B) 1340.0
(C) 1700.0
(D) 1560.0

Correct Answer: Option (A) : 1440.0

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (1440.0).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q185. What will be the output of the following pseudocode?

```
What is the output of the following C code? int f(int x,
int *py, int **ppz) { int y, z; **ppz += 1; z = **ppz;
*py += 2; y = *py; x += 3; return x + y + z; } void
main() { int c, *b, **a; c = 4; b = &c; a = &b;
printf("%d ", f(c, b, a)); return 0; }
```

(A) 18
(B) 19
(C) 21
(D) 22

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (19).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q186. What will be the output of the following pseudocode?

```
What is the output of the following C code? #include
<stdio.h> int main() { int a = 12; void *ptr = (int *)&a;
printf("%d", *ptr); getchar(); return 0; }
```

(A) 12
(B) Compiler Error
(C) Run Time Error
(D) 0

Correct Answer: Option (B) : Compiler Error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compiler Error).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q187. What will be the output of the following pseudocode?

```
Predict the output of the following pseudocode. Integer
x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3
& 3) && z + (2 ^ 2)) x = x - 2 y = x Else x = z y = y ^ 2
End if Print x + y + z
```

(A) 2
(B) 11
(C) 3
(D) 0

Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (3).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q188. What will be the output of the following pseudocode?

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x)
return 0; }
```

(A) 0
(B) 1
(C) 5
(D) Compilation Error

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q189. What will be the output of the following pseudocode?

```
Q1.) Integer p, q, r Set p = 0, q = 2, r = 9 r = 7 + p q = q + r for (each r from 4 to 7) p = (p + p) & q if ((p + q) < (r - p) || 8 < p) p = (p + 2) + q Jump out of the loop End if End for Print p + q
```

(A) 5
(B) 9
(C) 19
(D) 20

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (9).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q190. What will be the output of the following pseudocode?

```
Q2.) #include<iostream> using namespace std; int main() { int a = 3, b = 2, c = 5; int result = (a++ << 2) + (--b << 1) - (c-- >> 1) + (a & b); cout << result; return 0; }
```

(A) 12
(B) 14
(C) 10
(D) 16

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (12).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q191. What will be the output of the following pseudocode?

```
Q3.) Integer value, n Set value = 1, n = 45 while (value less than equal to n) value = value << 1 end loop Print value
```

(A) 16
(B) 32
(C) 64
(D) 36

Correct Answer: Option (C) : 64

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (64).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q192. What will be the output of the following pseudocode?

```
Q4.) Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x -2 y = x Else x = z y = y ^ 2 End if Print x + y + z
```

(A) 2
(B) 11
(C) 0
(D) 3

Correct Answer: Option (D) : 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (3).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q193. What will be the output of the following pseudocode?

```
Q5. What is the output int main() { int x = 10; int *y, **z; y = &x; z = &y; **z=7; print(x,*y,**z) return 0; }
```

(A) 10 10 10
(B) 10 7 7
(C) 7 7 7
(D) 10 10 7

Correct Answer: Option (C) : 7 7 7

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (7 7 7).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q194. What will be the output of the following pseudocode?

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value = p MOD q p = q q = value End while return p End function solve()
```

(A) 3
(B) 1
(C) 2
(D) 0

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q195. What will be the output of the following pseudocode?

```
Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number * something(number-1); } something(4);
```

(A) 12
(B) 24
(C) 1
(D) 20

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (24).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q196. Q8. What will be the output of the following pseudocode? int a , b ,c Set a=0,b=1,c=2 If (b^c || a&b || a>>1) c=9 a =b+c else c =1 a=a+c End if Print a+b+c

(A) 20

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q197. What will be the output of the following pseudocode?

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q198. What will be the output of the following pseudocode?

```
Q10. What is the output of following Integer value, n Set value = 32, n = 1 while(value >= n) value = value >> 1 end loop Print value
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q199. What will be the output of the following pseudocode?

```
What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print(“True”); Else Print(“False”); return 0; }
```

(A) True
(B) False
(C) Error
(D) -1

Correct Answer: Option (B) : False

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (False).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q200. What will be the output of the following pseudocode?

```
What is the output of following code: Int main(){ Int n=(1<<2+3<<4) Print(n); Return 0; }
```

(A) 112
(B) 512
(C) 52
(D) 0

Correct Answer: Option (B) : 512

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (512).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q201. What will be the output of the following pseudocode?

```
What is the output of following: Integer x, y Set x = 15, y = 12 y = x – 1 do{ Print x x = y + (x – 2) } while(x < 40) end do while
```

(A) 14 26
(B) 15 27 39
(C) 27 39
(D) 15

Correct Answer: Option (B) : 15 27 39

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (15 27 39).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q202. What will be the output of the following pseudocode?

```
What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If End Function
```

(A) 1 2 3 4 5
(B) 5 4 3 2 1
(C) Stack overflow
(D) -1

Correct Answer: Option (C) : Stack overflow

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (Stack overflow).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q203. What will be the output of the following pseudocode?

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }
```

(A) 0
(B) 1
(C) 5
(D) Compilation Error

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q204. What will be the output of the following pseudocode?

```
1. Integer a = 5, b = 9 2. a = a ^ b 3. b = a ^ b 4. a = a ^ b 5. if (a == 9 && b == 5) 6. Print "Swapped" 7. else 8. Print "Not swapped"
```

(A) Line 2
(B) Line 3
(C) Line 4
(D) No error

Correct Answer: Option (D) : No error

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (No error).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q205. What will be the output of the following pseudocode?

```
int main() { int x = 9, y = 2, z = 6; int a = x & y | z;
printf("%d", a); return 0; }
```

(A) Error
(B) 2
(C) 3
(D) 6

Correct Answer: Option (D) : 6

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (6).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q206. What will be the output of the following pseudocode?

```
Integer p, q, r Set p = 4, q = 2, r = 5 p = (q + p) + p
if ((q + p) < (r - q) && 3 > p) then p = (q + r) + p if
(((p + r) + (r + q)) <= (p + r + q)) then r = q + p else
r = (p + r) + p end if p = (9 & 6) + q else q = r & r if
((q - 5) < (5 - q)) then r = (6 + 6) & q q = (q + 7) + r
end if p = (1 ^ 11) + q end if Print p + q + r
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q207. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c >
b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c <
a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a
Print a + b + c
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q208. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 1, b = 6, c = 7 b = (a & b) + b
for (each c from 3 to 5) { b = (a + a) ^ a b = (12 + 5) +
a } a = (a ^ 9) + a Print a + b
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q209. What will be the output of the following pseudocode?

```
What is printed?
int f(int n) {
        if (n <= 1) return n;
        return f(n-1) + 2 * f(n-2);
}
print(f(6))
```

(A) 15
(B) 21
(C) 27
(D) 31

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q210. What will be the output of the following pseudocode?

```
What is printed?
x = 45; count = 0;
while (x != 0) {
        x = x & (x - 1);
        count++;
}
print(count ^ (45 >> 2))
```

(A) 4
(B) 11
(C) 15
(D) 7

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q211. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = i to 5
                if ((i + j) % 3 == 0) continue
                if (j == 5) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 38
(B) 42
(C) 46
(D) 50

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q212. What will be the output of the following pseudocode?

```
A stack is used (operands are evaluated left to right).
What is printed?
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
```

(A) 14
(B) 16
(C) 19
(D) 22

Correct Answer: Option (A) : 14

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (14).

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q213. What will be the output of the following pseudocode?

```
What is the final content of arr? (indexing starts at 0;
updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
        arr[i] = arr[i] + arr[arr[i] % 4]
end for
```

(A) [8, 2, 5, 7]
(B) [8, 2, 5, 6]
(C) [8, 5, 5, 7]
(D) [5, 2, 5, 7]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q214. What will be the output of the following pseudocode?

```
What is printed?
int f(int n) {
        if (n == 0) return 1;
        if (n == 1) return 2;
        return f(n-1) + f(n-2) * 2;
}
print(f(5))
```

(A) 26
(B) 30
(C) 32
(D) 34

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q215. What will be the output of the following pseudocode?

```
What is printed? (x and y are unsigned integers)
x = 58; y = 0;
while (x > 0) {
        y = (y << 1) | (x & 1);
        x = x >> 1;
}
print(y)
```

(A) 23
(B) 29
(C) 37
(D) 41

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q216. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 4
        for j = 1 to 4
                if (i == j) continue
                if (i + j > 6) break
                s = s + (i * j)
        end for
end for
print(s)
```

(A) 40
(B) 44
(C) 46
(D) 48

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q217. What will be the output of the following pseudocode?

```
A queue is used (FIFO). What is printed?
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
```

(A) 10
(B) 16
(C) 18
(D) 22

Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (10).

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q218. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]
arr[2] = arr[1]
arr[1] = temp
```

(A) [5, 1, 2, 4, 3]
(B) [5, 1, 3, 4, 2]
(C) [5, 2, 1, 4, 3]
(D) [5, 1, 2, 4, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q219. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 40
(B) 46
(C) 52
(D) 44

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q220. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 37
(B) 41
(C) 45
(D) 39

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q221. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 130
(B) 124
(C) 136
(D) 118

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q222. What will be the output of the following pseudocode?

```
A stack is used (LIFO). What is printed?
```

(A) 39
(B) 33
(C) 42
(D) 30

Correct Answer: Option (A) : 39

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (39).

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q223. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
```

(A) [60, 40, 30, 20, 50, 10]
(B) [60, 20, 30, 40, 50, 10]
(C) [10, 40, 30, 20, 50, 60]
(D) [60, 40, 50, 20, 30, 10]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q224. What will be the output of the following pseudocode?

```
What is printed? (% is the modulo operator)
int gcd(int a, int b) {
        if (b == 0) return a;
        return gcd(b, a % b);
}
print(gcd(252, 105))
```

(A) 21
(B) 42
(C) 7
(D) 14

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q225. What will be the output of the following pseudocode?

```
What is printed? (^=XOR, |=OR, &=AND, >>=right shift)
a = 13; b = 25;
c = (a ^ b) & (a | b);
d = c >> 2;
print(c + d)
```

(A) 25
(B) 29
(C) 20
(D) 33

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q226. How many swaps are performed in total?

```
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
        for j = 0 to 4 - i
                if (arr[j] > arr[j+1])
                        temp = arr[j]
                        arr[j] = arr[j+1]
                        arr[j+1] = temp
                        swaps = swaps + 1
                end if
        end for
end for
print(swaps)
```

(A) 7
(B) 8
(C) 6
(D) 9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q227. What will be the output of the following pseudocode?

```
This algorithm reverses the first k elements of a queue
using a stack. What is printed?
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
        push(stack, dequeue(queue))
end for
while (stack is not empty)
        enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
        enqueue(queue, dequeue(queue))
end for
print(queue)
```

(A) [3, 2, 1, 4, 5]
(B) [1, 2, 3, 4, 5]
(C) [5, 4, 3, 2, 1]
(D) [3, 2, 1, 5, 4]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q228. What will be the output of the following pseudocode?

```
This algorithm moves all non-zero elements to the front,
preserving order, and fills the rest with zeroes.
What is the final arr?
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
        if (arr[i] != 0)
                arr[pos] = arr[i]
                pos = pos + 1
        end if
end for
while (pos <= 6)
        arr[pos] = 0
        pos = pos + 1
end while
print(arr)
```

(A) [3, 4, 5, 2, 0, 0, 0]
(B) [3, 0, 4, 0, 5, 0, 2]
(C) [3, 4, 5, 2, 0, 0, 2]
(D) [0, 0, 0, 3, 4, 5, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q229. What will be the output of the following pseudocode?

```
What is printed? (This computes the number of grid paths
from one corner to another, moving only
right or down.)
int f(int n, int m) {
        if (n == 0 || m == 0) return 1;
        return f(n-1, m) + f(n, m-1);
}
print(f(3, 2))
```

(A) 10
(B) 8
(C) 12
(D) 9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q230. What will be the output of the following pseudocode?

```
What is printed? (>>=right shift, <<=left shift, &=AND,
^=XOR)
x = 0;
for i = 0 to 7
        if (((45 >> i) & 1) == 1)
                x = x + (1 << (7 - i))
        end if
end for
y = x ^ 45
print(y)
```

(A) 153
(B) 180
(C) 135
(D) 171

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q231. This is a binary search for target = 50 in a sorted array. How many comparisons are made before

```
the loop ends (the target is not present)?
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low <= high)
        mid = (low + high) / 2
        comparisons = comparisons + 1
        if (arr[mid] == target)
                break
        else if (arr[mid] < target)
                low = mid + 1
        else
                high = mid - 1
        end if
end while
print(comparisons)
```

(A) 4
(B) 3
(C) 5
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q232. What will be the output of the following pseudocode?

```
This stack trace evaluates a postfix-style expression.
What is printed?
push(6); push(2); push(3);
c = pop(); b = pop(); push(b + c);
d = pop(); e = pop(); push(e - d);
push(3); push(8); push(2);
f = pop(); g = pop(); push(g / f);
h = pop(); i = pop(); push(i + h);
j = pop(); k = pop(); push(k * j);
print(pop())
```

(A) 7
(B) 10
(C) 14
(D) 4

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q233. What will be the output of the following pseudocode?

```
This is Kadane's algorithm for maximum subarray sum. What
is printed?
arr = [-2, 3, -1, 5, -3, 4, -2, 6]
maxSoFar = arr[0]; maxEndingHere = arr[0]
for i = 1 to 7
        maxEndingHere = max(maxEndingHere + arr[i],
arr[i])
        maxSoFar = max(maxSoFar, maxEndingHere)
end for
print(maxSoFar)
```

(A) 12
(B) 10
(C) 9
(D) 15

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q234. What will be the output of the following pseudocode?

```
What is printed? (/ is integer division, % is modulo)
int f(int n) {
        if (n <= 1) return n;
        if (n % 2 == 0) return f(n / 2);
        return f(n - 1) + f(n + 1);
}
print(f(11))
```

(A) 5
(B) 4
(C) 6
(D) 3

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q235. What will be the output of the following pseudocode?

```
What is printed? (^=XOR, >>=right shift, &=AND)
n = 13
g = n ^ (n >> 1)
print(g + (g & 5))
```

(A) 12
(B) 11
(C) 16
(D) 14

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q236. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = 1 to 5
                if ((i + j) % 4 == 0) continue
                if (j > i + 1) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 142
(B) 138
(C) 146
(D) 150

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 6]

Topic: *Infosys SE College Mock Test*

Q237. The following sorts a stack using a temporary stack (the top of a stack is its last element). How many

```
times is an element moved from tmp back to st?
st = [3, 1, 4, 2]; tmp = []; moves = 0
while (st is not empty)
        t = pop(st)
        while (tmp is not empty and top(tmp) > t)
                push(st, pop(tmp))
                moves = moves + 1
        end while
        push(tmp, t)
end while
print(moves)
```

(A) 3
(B) 2
(C) 4
(D) 5

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 6]

Topic: *Infosys SE College Mock Test*

Q238. What will be the output of the following pseudocode?

```
What is printed? (dynamic programming for the minimum
number of coins; INF is a very large number)
coins = [1, 3, 4]
dp[0] = 0; dp[1..6] = INF
for i = 1 to 6
        for each c in coins
                if (i >= c)
                        dp[i] = min(dp[i], dp[i - c] + 1)
                end if
        end for
end for
print(dp[6])
```

(A) 2
(B) 3
(C) 4
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .